

Infectious Disease Review

Integrated Cases, Diagnostic Algorithms, and Graph-RAG Extraction Map

This capstone chapter is designed for high-density retrieval and clinical reasoning. It links fever syndromes, source identification, microbiology, empiric therapy, complications, and diagnostic traps across the infectious disease section. The goal is not to memorize every organism in isolation, but to recognize how host factors, exposure history, immune status, anatomy, cultures, imaging, and urgency determine the next step.

How to use this chapter

Read each syndrome as an illness script: host risk + exposure + time course + anatomic source + key test + empiric treatment + dangerous complication. For Graph-RAG extraction, each heading can become a concept node, while table rows and arrows define edge candidates such as causes, predisposes-to, mimics, treats, contraindicates, complicates, and diagnostic-test-for.

Core Diagnostic Framework

The safest infectious disease workflow starts with acuity, then source, then host, then organism. A febrile patient with hypotension, hypoxia, altered mental status, lactate elevation, neutropenia, meningismus, necrotizing soft tissue findings, or rapidly progressive rash should not wait for complete microbiologic certainty. Cultures matter, but source control and empiric therapy are time-sensitive when organ dysfunction is present.

Step	Clinical question	High-yield anchors	Common trap
1. Stabilize	Is this sepsis, shock, meningitis, necrotizing infection, neutropenic fever, or toxic shock?	Vital signs, mental status, lactate, oxygenation, urine output, hypotension, rapidly progressive pain/rash.	Waiting for culture confirmation before treating an unstable patient.
2. Localize source	Which anatomic compartment is infected?	Lung, urine, CNS, abdomen, skin/soft tissue, bone/joint, vascular catheter, blood, endocardium.	Treating fever as a diagnosis rather than a clue to a source.
3. Identify host risks	What immune or structural factor changes likely organisms?	Neutropenia, HIV/CD4 count, transplant, splenectomy, diabetes, cirrhosis, hardware, catheter, aspiration risk, travel.	Using a standard outpatient regimen in a high-risk host.
4. Obtain focused tests	Which tests will change management?	Blood cultures, urine studies, CXR/CT, LP, wound/joint aspiration, stool testing, viral testing, serology when appropriate.	Ordering broad panels without asking whether the result will change treatment.
5. Start empiric therapy	What must be covered immediately?	Likely pathogens plus resistant organisms if risk factors exist. Include source control when needed.	Antibiotics without drainage, catheter removal, debridement, or device management.
6. Narrow and monitor	Can therapy be de-escalated?	Culture results, susceptibilities, clinical response, repeat imaging only when response is poor.	Continuing vancomycin/piperacillin-tazobactam by inertia.

Fever, Sepsis, and Culture Logic

Fever is a regulated rise in hypothalamic set point, whereas hyperthermia reflects failed heat dissipation or excess heat production. Infectious fever often responds to antipyretics, but lack of response does not prove noninfectious disease. Fever of unknown origin is a diagnosis of exclusion after persistent fever, adequate time, and an unrevealing initial workup. The most common broad categories are infection, malignancy, inflammatory disease, and drug fever.

Sepsis is infection with life-threatening organ dysfunction. The first therapeutic move is not a perfect organism guess; it is recognition of instability, rapid source search, cultures when they do not delay treatment, early antimicrobials, fluids/vasopressors as indicated, and source control. Blood cultures are most valuable before antibiotics in suspected bacteremia, endocarditis, catheter-related infection, meningitis, severe pneumonia, pyelonephritis with sepsis, and neutropenic fever.

Scenario	Immediate action	Core tests	Empiric concept
Undifferentiated septic shock	Airway/oxygen, IV access, fluids if hypoperfused, vasopressors if persistent hypotension, source control.	Blood cultures x2, lactate, CBC/CMP, UA/culture, CXR, source-directed imaging.	Broad coverage for likely source; narrow once data return.
Meningitis concern	Do not delay antibiotics for CT if unstable. Add dexamethasone when bacterial meningitis likely and before/with first dose.	Blood cultures, LP if safe, CT before LP if focal deficits, papilledema, seizure, immunocompromise, altered consciousness.	Ceftriaxone/cefotaxime + vancomycin; add ampicillin for Listeria risk.
Neutropenic fever	Medical emergency even if patient looks well.	Blood cultures from peripheral and catheter sites, urine, CXR if symptoms, inspect skin/mucosa/catheter/perianal area.	Antipseudomonal beta-lactam; add vancomycin only for selected indications.
Catheter-related sepsis	Remove catheter if severe sepsis, S. aureus, Candida, tunnel infection, persistent bacteremia, or unstable host.	Peripheral and line cultures; catheter tip culture if removed.	Vancomycin plus gram-negative coverage depending severity/local risks; tailor to organism.
Toxic shock syndrome	Hemodynamic support and immediate source removal.	Cultures may be negative; diagnosis is clinical.	Antistaphylococcal/MRSA coverage plus toxin suppression such as clindamycin.

Respiratory Infection Algorithms

Respiratory infections are sorted by setting, host, radiographic pattern, and aspiration/ventilator exposure. Community-acquired pneumonia typically arises from microaspiration of oropharyngeal flora and is divided into typical and atypical patterns. Hospital-acquired and ventilator-associated pneumonia require attention to gram-negative rods, *Pseudomonas*, and MRSA risk. Lung abscess is usually an aspiration-driven anaerobic infection with cavitation and foul sputum, and tuberculosis is a chronic airborne infection with granulomatous host response and risk of reactivation.

Syndrome	Typical host/exposure	Clues	Key tests	Treatment direction
Typical CAP	Community patient; older age, COPD, alcoholism, asplenia increase severity risk.	Acute fever/chills, productive cough, pleuritic pain, lobar consolidation.	CXR; sputum/blood cultures if hospitalized or severe.	Outpatient vs inpatient regimen based on severity/comorbidity; cover pneumococcus and atypicals as indicated.
Atypical pneumonia	Younger adults, outbreaks, animal/water exposure depending organism.	Dry cough, headache, myalgias, diffuse infiltrates, pulse-temperature dissociation in some cases.	CXR; targeted tests for Legionella/influenza when severe or epidemiologically suggested.	Macrolide or doxycycline concept; fluoroquinolone in selected adults.
HAP/VAP	Hospitalization >48 h or mechanical ventilation.	New infiltrate plus fever, purulent secretions, leukocytosis, worsening oxygenation.	Respiratory culture, blood cultures; assess local antibiogram.	Antipseudomonal beta-lactam +/- MRSA coverage; de-escalate.
Lung abscess	Aspiration risk, poor dentition, seizures, alcohol use, dysphagia.	Indolent fever, foul sputum, weight loss, cavity with air-fluid level.	CXR/CT; bronchoscopy if obstruction or atypical course.	Anaerobic and oral flora coverage; prolonged therapy; drainage rarely.
Tuberculosis	Exposure, incarceration, homelessness, HIV, immunosuppression, endemic travel.	Night sweats, weight loss, cough, hemoptysis; upper-lobe cavitation or miliary disease.	AFB smear/culture, NAAT, CXR; TST/IGRA for infection not active disease.	RIPE therapy for active TB; public health isolation and contact evaluation.
Influenza	Winter season, abrupt systemic illness, high-risk host.	Fever, myalgias, cough, sore throat; bacterial superinfection possible.	Rapid molecular testing when result changes therapy/isolation.	Supportive care; oseltamivir/zanamivir most useful early or in high-risk/severe illness.

CNS Infection Algorithm

Meningitis is a time-sensitive diagnosis. Bacterial meningitis classically produces fever, headache, neck stiffness, photophobia, altered mental status, high CSF neutrophils, elevated protein, and low glucose. Viral meningitis tends to have lymphocytic predominance, normal glucose, and milder course. The most dangerous mistake is delaying antibiotics for imaging or lumbar puncture when the patient is unstable or bacterial meningitis is likely.

Finding	Bacterial meningitis	Aseptic/viral meningitis	Cryptococcal/fungal concept
Host clue	Extremes of age, otitis/sinusitis, asplenia, complement deficiency, basilar skull fracture.	Enterovirus, HSV/VZV, mild systemic viral prodrome.	Advanced HIV, transplant, chronic immunosuppression.
CSF cells	High WBC, PMN predominance.	WBC elevation with lymphocyte predominance.	Variable WBC; may be modest in severe immunosuppression.

Finding	Bacterial meningitis	Aseptic/viral meningitis	Cryptococcal/fungal concept
CSF glucose	Low.	Normal.	Often low or normal-low.
CSF protein	High.	Normal to moderately high.	High.
Test emphasis	Gram stain/culture, blood cultures, PCR adjuncts.	PCR for HSV/enterovirus when relevant.	Cryptococcal antigen, fungal culture, opening pressure.
Treatment principle	Immediate bactericidal therapy; add dexamethasone when indicated.	Supportive except HSV/VZV concern requires acyclovir.	Amphotericin B + flucytosine induction, then fluconazole consolidation.

GI, Hepatitis, C. difficile, and Intra-Abdominal Infections

GI infectious reasoning separates mucosal toxin-mediated illness, invasive inflammatory diarrhea, antibiotic-associated C. difficile, hepatotropic viral syndromes, and deep intra-abdominal collections. Intra-abdominal infections are commonly polymicrobial and often require drainage or operative source control in addition to antibiotics. Viral hepatitis is best decoded by transmission route, chronicity risk, serologic pattern, and risk of cirrhosis/hepatocellular carcinoma.

Condition	Illness script	Key diagnostic anchor	Treatment/source control concept
Viral hepatitis A/E	Fecal-oral acute hepatitis; usually no chronic infection.	IgM anti-HAV; HEV testing when epidemiology fits.	Supportive; prevention through sanitation/vaccination where applicable.
HBV	Parenteral/sexual/perinatal; can be acute, chronic, carrier, or fulminant.	HBsAg infection; anti-HBc IgM acute/window; HBeAg high infectivity.	Vaccination prevents; chronic therapy depends viral load, inflammation/fibrosis, comorbidity.
HCV	Blood exposure; often silent acute phase; chronic infection common.	HCV antibody exposure; HCV RNA active infection.	Direct-acting antivirals cure most patients; stage fibrosis and screen complications.
C. difficile colitis	Antibiotic exposure, watery diarrhea, cramping, leukocytosis; severe disease can cause ileus/toxic megacolon.	Stool toxin/NAAT in symptomatic patient.	Stop inciting antibiotic if possible; oral vancomycin or fidaxomicin; surgery for fulminant complications.
Intra-abdominal abscess	Persistent fever or sepsis after perforation, pancreatitis, pelvic infection, surgery, or bowel disease.	CT/ultrasound collection; culture if drained.	Drainage plus gram-negative, enterococcal, and anaerobic coverage.
Botulism	Descending symmetric flaccid paralysis after preformed toxin exposure or wound contamination.	Toxin assay; diagnosis often clinical.	Airway monitoring, antitoxin, wound care if wound botulism.

GU and STI Infection Algorithms

Urinary infection reasoning is dominated by anatomy and host factors: uncomplicated cystitis is a bladder infection in a nonpregnant woman without structural disease, while complicated UTI includes male sex in many contexts, pregnancy, catheter, obstruction, immunosuppression, renal impairment, diabetes, stones, or upper-tract involvement. Pyelonephritis adds fever/flank pain/CVA tenderness and may progress to bacteremia or shock. Prostatitis is suggested by pelvic/perineal pain, obstructive urinary symptoms, fever in acute disease, and a tender boggy prostate.

Syndrome	Clues	Testing	Treatment concept
Uncomplicated cystitis	Dysuria, frequency, urgency, suprapubic pain; fever usually absent.	UA leukocyte esterase/nitrite; culture if recurrent, atypical, pregnancy, resistance risk, or treatment failure.	Short oral regimen such as nitrofurantoin, TMP-SMX when resistance acceptable, or fosfomycin.
Pyelonephritis	Fever, flank pain, CVA tenderness, nausea/vomiting, pyuria; may have bacteremia.	UA and urine culture; blood cultures if ill/hospitalized; imaging if poor response or complicated features.	Oral therapy if stable; IV therapy and hospitalization if toxic, pregnant, unable PO, or complicated.
Acute bacterial prostatitis	Fever/chills, dysuria, urinary retention, perineal pain, tender boggy prostate.	Urine culture; blood cultures if toxic. Avoid vigorous massage.	Fluoroquinolone or TMP-SMX if outpatient; IV therapy if severe. Longer duration than cystitis.
Chlamydia urethritis/cervicitis	Often asymptomatic; urethral/cervical discharge, dysuria, PID/epididymitis complications.	NAAT from urine or genital site.	Doxycycline or recommended alternative; treat partners.
Gonorrhea	Purulent urethritis/cervicitis, PID, epididymitis, disseminated arthritis/dermatitis.	NAAT; culture if treatment failure or resistance concern.	Ceftriaxone-based therapy; add chlamydia coverage if not excluded.

Syndrome	Clues	Testing	Treatment concept
Syphilis	Painless chancre, rash including palms/soles, neurologic/cardiovascular late disease.	Nontreponemal plus treponemal testing; CSF if neurologic/ocular disease.	Penicillin regimen by stage.

Skin, Soft Tissue, Bone, and Joint Infections

Skin and musculoskeletal infections are governed by depth. Cellulitis is dermal/subcutaneous inflammation; erysipelas is more superficial and sharply demarcated; abscess requires drainage; necrotizing fasciitis requires urgent surgical exploration; osteomyelitis requires long therapy and sometimes debridement; septic arthritis requires joint drainage plus antibiotics. Pain out of proportion, crepitus, bullae, anesthesia, systemic toxicity, rapid progression, or shock should trigger concern for necrotizing infection.

Syndrome	Distinguishing features	Likely organisms/exposures	Management anchor
Cellulitis	Erythema, warmth, swelling, tenderness; often unilateral.	Strep pyogenes, S. aureus; water exposure changes pathogen list.	Clinical diagnosis; cover streptococci; MRSA coverage if risk factors/purulence.
Erysipelas	Well-demarcated fiery red superficial lesion; lymphatic involvement.	Usually group A strep.	Penicillin/amoxicillin concept for uncomplicated cases.
Abscess	Fluctuant collection; purulence.	S. aureus including MRSA.	Incision and drainage is primary; antibiotics if systemic signs, immunocompromise, extensive disease.
Necrotizing fasciitis	Pain out of proportion, rapid progression, toxicity, bullae, crepitus, anesthesia.	Group A strep, polymicrobial anaerobes/gram-negatives, clostridia.	Emergency surgical exploration/debridement plus broad antibiotics and toxin suppression.
Osteomyelitis	Bone pain, chronic wound, diabetic foot ulcer, open fracture, hardware, vertebral pain.	S. aureus common; Pseudomonas in puncture through shoe; Salmonella in sickle cell.	MRI sensitive; bone biopsy/culture best; 4-6+ weeks therapy; debridement if necrotic bone.
Septic arthritis	Hot swollen painful joint with very limited active/passive ROM.	S. aureus overall; N. gonorrhoeae in sexually active young adults.	Arthrocentesis before antibiotics if possible; drainage plus antibiotics.

Travel, Zoonotic, Fungal, and Parasitic Patterns

Exposure history often supplies the diagnosis before the laboratory does. Tick exposure plus erythema migrans strongly suggests Lyme disease; fever with rash beginning on wrists/ankles and moving centrally suggests RMSF; periodic fever after travel raises malaria; animal bite with neurologic symptoms raises rabies concern; advanced HIV with subacute meningitis suggests cryptococcus; neutropenia with persistent fever raises invasive mold concern.

Exposure/host clue	Likely syndrome	Clinical anchor	Diagnostic/treatment anchor
Ixodes tick, Northeast/Upper Midwest, expanding targetoid rash	Lyme disease	Erythema migrans, facial palsy, meningitis, AV block, migratory arthritis.	Clinical diagnosis if classic EM; doxycycline/amoxicillin/cefuroxime depending stage and host.
Tick exposure, fever, severe headache, rash wrists/ankles/palms/soles	RMSF	Rash spreads centrally; thrombocytopenia and transaminitis may occur.	Treat immediately with doxycycline; do not wait for serology.
Travel to endemic region, fever/chills, hemolysis risk	Malaria	P. falciparum is most severe; smear with Giemsa.	Thick/thin smears or rapid test; therapy depends species, severity, resistance, pregnancy.
Dog/bat/raccoon exposure	Rabies	Hydrophobia, encephalitis, almost universally fatal once symptomatic.	Postexposure wound care, immune globulin, vaccine series when indicated.
Broad antibiotics/diabetes/HIV/catheter/TPN	Candida	Thrush, vaginitis, intertrigo, candidemia in high-risk host.	KOH/culture; remove catheter for candidemia; azole/echinocandin/amphotericin by severity.
Neutropenia/transplant; lung nodules; hemoptysis	Aspergillus	ABPA, aspergilloma, invasive angioinvasive disease.	CT, galactomannan/culture; voriconazole-based treatment for invasive disease.
Advanced AIDS, pigeon exposure, subacute meningitis	Cryptococcus	Headache, fever, elevated opening pressure, low inflammation possible.	CSF cryptococcal antigen; amphotericin B + flucytosine induction.

Special Host Patterns

Host status modifies the organism list and the threshold for admission. A patient with normal immunity and mild cystitis is different from a neutropenic patient with fever, a transplant recipient with pulmonary nodules, a cirrhotic patient with abdominal

pain/ascites, or an advanced HIV patient with meningitis. In immunocompromised hosts, fever may be muted, inflammatory signs may be absent, and radiographs may lag behind disease.

Host factor	What changes?	Classic infection patterns	Clinical trap
Neutropenia	Reduced inflammatory response; fever may be only sign.	Gram-negative bacteremia, <i>Pseudomonas</i> , cellulitis, pneumonia, candidemia/mold if prolonged.	Reassuring exam despite severe infection.
Advanced HIV	CD4-specific opportunistic infections.	PJP, toxoplasmosis, cryptococcus, CMV, MAC, TB, candidiasis.	Missing OIs because routine WBC response is absent.
Asplenia	Poor clearance of encapsulated organisms.	<i>S. pneumoniae</i> , <i>H. influenzae</i> , <i>N. meningitidis</i> , severe babesiosis.	Underestimating rapidly fatal sepsis risk.
Diabetes	Vascular disease and immune dysfunction.	UTI/pyelo, emphysematous infection, diabetic foot, mucormycosis in DKA, necrotizing infection.	Treating foot ulcer without probing/debriding/evaluating bone.
Cirrhosis/ascites	Immune dysfunction, bacterial translocation.	SBP, gram-negative bacteremia, <i>Vibrio</i> risk with seawater/raw oysters.	Missing subtle SBP; fever may be absent.
Indwelling devices/hardware	Biofilm and persistent infection.	Catheter sepsis, prosthetic joint infection, pacemaker/endocarditis, shunt infection.	Antibiotics without device/source control.

High-Yield Diagnostic Algorithms

Fever + rash

Start with acuity and rash morphology. Petechiae/purpura plus toxicity: meningococcemia, RMSF, DIC, TTP, endocarditis, viral hemorrhagic fever depending exposure. Diffuse erythematous rash plus shock: toxic shock, scarlet fever, drug reaction. Vesicular lesions: varicella-zoster or HSV. Targetoid expanding lesion after tick exposure: Lyme. Palms/soles involvement: RMSF, syphilis, meningococcemia, hand-foot-mouth, drug eruption.

Fever + neurologic symptoms

Meningismus suggests meningitis. Altered mental status without neck stiffness suggests encephalitis, sepsis, metabolic disease, toxidrome, or abscess. Focal deficits require CT/MRI before LP. In advanced HIV, consider cryptococcus, toxoplasmosis, TB meningitis, lymphoma, and PML. Treat suspected bacterial meningitis promptly.

Fever + pulmonary infiltrate

Ask community vs hospital vs ventilator, aspiration risk, immune status, TB risk, travel/fungal exposure, and severity. Lobar consolidation supports typical bacterial pneumonia; diffuse interstitial pattern suggests atypical/viral/PJP; cavitation suggests abscess, TB, *S. aureus*, *Klebsiella*, fungal disease, malignancy, or septic emboli.

Fever + diarrhea

Watery nonbloody diarrhea after antibiotics suggests *C. difficile*. Bloody diarrhea with fever suggests invasive bacterial colitis. Descending paralysis after home-canned food suggests botulism. Severe dehydration or hemolytic uremic syndrome clues change urgency. Avoid antimotility agents when invasive colitis or *C. difficile* is suspected.

Fever + joint pain

A single hot joint is septic arthritis until proven otherwise. Arthrocentesis is the key test. Polyarticular migratory symptoms after STI exposure can be disseminated gonorrhea. Reactive arthritis follows GU/GI infection. Crystal disease can mimic infection; crystals do not exclude infection.

Fever + central line

Inspect line but do not rely on local findings. Obtain peripheral and line cultures. Remove catheter when severe sepsis, persistent bacteremia, *S. aureus*, *Candida*, tunnel infection, or endovascular complications are suspected.

Integrated Cases

Case 1 - Febrile cough with pleuritic pain

A 67-year-old with COPD has fever, productive cough, tachypnea, pleuritic pain, and right lower-lobe consolidation. This is community-acquired pneumonia until proven otherwise. Assess severity and oxygenation, obtain CXR, consider blood/sputum cultures if hospitalized, and start empiric CAP therapy. If the patient was recently hospitalized or ventilated, broaden for resistant

gram-negative rods and MRSA risk.

Case 2 - Antibiotic-associated diarrhea

A hospitalized patient treated with clindamycin develops watery diarrhea, leukocytosis, and abdominal cramping. *C. difficile* is likely. Test only symptomatic patients, stop unnecessary antibiotics, give vancomycin or fidaxomicin for initial disease, and monitor for ileus, toxic megacolon, shock, and need for surgery.

Case 3 - Hot swollen knee

A patient presents with fever and a swollen painful knee with limited passive range of motion. Septic arthritis is an emergency. Aspirate the joint for cell count, Gram stain, culture, and crystals, then start empiric antibiotics. Drainage is required because antibiotics alone do not clear a purulent joint space reliably.

Case 4 - Neutropenic fever after chemotherapy

A patient 8 days after chemotherapy has ANC 300 and temperature 38.4 C. Even if the exam is benign, treat as neutropenic fever. Draw cultures, inspect mucosa/skin/catheter/perianal area, obtain focused imaging based on symptoms, and begin antipseudomonal beta-lactam therapy promptly. Add antifungal therapy if fever persists despite antibacterial therapy and neutropenia is prolonged.

Case 5 - Tick exposure with rash and headache

A camper has fever, severe headache, thrombocytopenia, transaminitis, and rash beginning on wrists and ankles. RMSF should be treated immediately with doxycycline. Early serology can be negative, and delay increases mortality.

Case 6 - HIV patient with subacute headache

An advanced HIV patient has progressive headache, fever, and confusion. Cryptococcal meningitis is high on the differential. LP with opening pressure and cryptococcal antigen testing are central. Treatment requires amphotericin B plus flucytosine induction and management of elevated intracranial pressure.

Case 7 - Severe soft tissue pain

A diabetic patient has thigh pain out of proportion to exam, fever, tachycardia, and evolving dusky skin. Necrotizing fasciitis must be presumed. Do not wait for imaging if unstable. Broad antibiotics and immediate surgical exploration/debridement are lifesaving.

Case 8 - Prolonged unexplained fever

A patient has fevers for several weeks after a nondiagnostic initial evaluation. FUO workup should revisit medications, travel, animal exposures, immune status, indwelling devices, occult abscess, endocarditis, malignancy, rheumatologic disease, and granulomatous disease. Empiric antibiotics or steroids can obscure the diagnosis unless the patient is unstable.

Common Pitfalls

- Normal early CXR does not exclude pneumonia in dehydration, neutropenia, PCP/PJP, or very early disease.
- Absence of fever does not exclude severe infection in older adults, immunosuppressed patients, or patients taking antipyretics/steroids.
- A positive urine culture without symptoms is usually asymptomatic bacteriuria, not UTI, except in pregnancy or before selected urologic procedures.
- Crystals in synovial fluid do not exclude septic arthritis.
- Antibiotics alone are inadequate for abscess, necrotizing fasciitis, empyema, many intra-abdominal abscesses, catheter sepsis, and infected necrotic bone.
- Do not delay treatment for suspected meningitis, RMSF, septic shock, necrotizing fasciitis, or neutropenic fever while waiting for confirmatory tests.
- Broad antibiotics should be narrowed once cultures and clinical response identify the pathogen and source.
- TB testing with TST/IGRA supports infection but does not distinguish latent from active disease; active disease requires microbiologic and radiographic evaluation.
- In *C. difficile*, testing formed stool or colonized patients can cause overdiagnosis.

- In toxic shock, cultures may be negative because toxin, not bloodstream bacterial burden, drives the syndrome.

Practice Questions

1. A 24-year-old develops fever, headache, and a rash on wrists and ankles 5 days after a hiking trip. Platelets are low and AST/ALT are mildly elevated. Best next step?

Answer: Start doxycycline immediately for suspected Rocky Mountain spotted fever. Do not wait for serologic confirmation.

2. A patient with leukemia has ANC 200 and a temperature of 38.5 C but no localizing symptoms. Best initial management?

Answer: Obtain cultures promptly and start antipseudomonal empiric antibacterial therapy. Benign exam is not reassuring in neutropenia.

3. A patient with a painful swollen knee has synovial WBC 80,000 with PMN predominance and negatively birefringent crystals. What is the trap?

Answer: Gout and septic arthritis can coexist. Continue culture-directed evaluation and treat infection if clinical suspicion remains.

4. A woman has dysuria and frequency but no fever, flank pain, pregnancy, or structural urinary disease. What syndrome is this?

Answer: Uncomplicated cystitis. Short-course oral therapy is appropriate; broad IV therapy is unnecessary.

5. A patient with advanced AIDS has subacute headache and elevated opening pressure on LP. India ink shows encapsulated yeast. Diagnosis and treatment?

Answer: Cryptococcal meningitis; amphotericin B plus flucytosine induction, followed by fluconazole consolidation, plus management of intracranial pressure.

6. A diabetic patient has rapidly progressive leg pain, fever, and skin anesthesia. Antibiotics are started. What critical intervention is still required?

Answer: Immediate surgical exploration and debridement for suspected necrotizing fasciitis.

7. A hospitalized patient has watery diarrhea after antibiotics and leukocytosis. Which treatment is preferred for an initial episode?

Answer: Oral vancomycin or fidaxomicin, with discontinuation of unnecessary inciting antibiotics when possible.

8. A patient has fever for 4 weeks, negative initial cultures, and no diagnosis after repeated evaluation. What broad categories frame FUO?

Answer: Infection, malignancy, inflammatory/autoimmune disease, drug fever, and miscellaneous causes such as granulomatous disease or thromboembolism.

Graph-RAG Extraction Map

The following map is designed for extraction into nodes and edges. Each concept should preserve syndrome, host context, diagnostic anchor, treatment anchor, and common mimics.

Node class	Example nodes	Important properties	Edge examples
Syndrome	Sepsis, FUO, CAP, meningitis, pyelonephritis, cellulitis, osteomyelitis, septic arthritis, neutropenic fever.	Urgency, source, host risk, hallmark findings, complications.	sepsis requires source control; meningitis treated-by ceftriaxone + vancomycin; neutropenic fever requires antipseudomonal therapy.
Pathogen	S. pneumoniae, S. aureus, Pseudomonas, N. meningitidis, M. tuberculosis, C. difficile, Candida, Aspergillus, Cryptococcus.	Gram stain, reservoir, exposure, resistance risk, preferred tests.	S. pneumoniae causes CAP/meningitis; C. difficile follows antibiotic exposure; Aspergillus risk-in neutropenia.

Node class	Example nodes	Important properties	Edge examples
Host factor	Neutropenia, HIV, asplenia, diabetes, cirrhosis, catheter, prosthetic joint, aspiration risk.	Immune defect, likely pathogens, altered presentation.	asplenia predisposes-to encapsulated bacteria; diabetes predisposes-to diabetic foot infection.
Diagnostic test	Blood culture, LP, CXR, CT, MRI, NAAT, serology, arthrocentesis, bone biopsy.	Sensitivity timing, specimen, when required, limitations.	arthrocentesis diagnoses septic arthritis; MRI detects osteomyelitis; IGRA supports TB infection.
Treatment	Empiric antibiotics, antivirals, antifungals, drainage, debridement, catheter removal, vaccination, prophylaxis.	Coverage target, timing, contraindication, duration concept.	necrotizing fasciitis requires debridement; abscess treated-by incision and drainage.
Complication	Shock, ARDS, empyema, toxic megacolon, renal failure, endocarditis, intracranial pressure, splenic rupture.	Syndrome link, severity, monitoring.	C. difficile complicates-to toxic megacolon; mono complicates-to splenic rupture.

Source Coverage Note

This integrated review synthesizes the uploaded infectious disease source sections: respiratory/TB/CNS infections; GI/GU/STI infections; skin/soft tissue/bone/joint infections; zoonotic, travel, fungal, and parasitic infections; and fever/sepsis/neutropenic fever/mononucleosis content. It is written as an original study resource rather than a reproduction of the references.